

ViPragSent: Implicit Sentiment, Sarcasm, and Figurative Language in Vietnamese Social Media

Anonymous ACL submission

Abstract

Vietnamese sentiment benchmarks such as UIT-VSFC, UIT-VSMEC and AIVIVN have driven steady progress on surface polarity, but they under-cover the pragmatic phenomena that dominate real Vietnamese social media: implicit sentiment, sarcasm, irony, idiomatic and figurative expressions, code-switching with English slang, and mocking. We present ViPragSent, a pragmatic sentiment analyser for Vietnamese social text. ViPragSent extends 8.4k Facebook, TikTok and YouTube comments with adjudicated pragmatic labels and trains shared encoders (PhoBERT, XLM-R) and small instruction language models (Sailor-7B, Vistral-7B) under a multi-task loss with auxiliary emotion and ordinary-sentiment heads and a chain-of-thought rationale-generation head used only during training. Across the six pragmatic phenomena, ViPragSent improves macro-F1 by +5.5 over a PhoBERT fine-tune, with +7.4 on sarcasm and +5.9 on implicit sentiment; ordinary sentiment is retained or improved on all three public datasets (+2.4 on UIT-VSMEC, +0.6 on UIT-VSFC, +0.5 on AIVIVN). Against the strongest baseline, a few-shot GPT-4o-mini prompt, the gain is +7.2 on sarcasm and +3.2 on implicit sentiment. Multi-task transfer and the explanation-augmented auxiliary each contribute roughly half of the gain; removing the rationale head costs 1.5 points and removing multi-task losses costs 6.4 points. Expected calibration error drops from 0.094 to 0.048. We release the new dataset, the trained models, the result JSON files, the figure-generation script, and a claim ledger that traces every numeric claim in the paper to a source file.

1 Introduction

Vietnamese sentiment analysis has grown around a small set of publicly available datasets: UIT-VSFC for student feedback (Nguyen et al., 2018b), UIT-VSMEC for seven-way emotion (Ho et al., 2020),

AIVIVN for e-commerce review polarity (AIVIVN Organisers, 2019), and a wider infrastructure of Vietnamese benchmarks for general language understanding, reading comprehension, lexical normalisation and dialect (Do et al., 2024; Tran et al., 2024; Nguyen et al., 2020b, 2025, 2024). Strong Vietnamese encoders — PhoBERT (Nguyen and Nguyen, 2020) and ViSoBERT (Nguyen et al., 2023) — now solve the headline 3-way polarity tasks almost saturated, with UIT-VSFC macro-F1 above 91% and AIVIVN above 90%.

A different picture appears when one reads through Vietnamese Facebook, TikTok, and YouTube comments: the dominant difficulty is not lexical polarity but *pragmatic interpretation*. A surface phrase such as “*hay lắm =)*” (“so very good =)”) is literally positive, but the trailing tilted face is a sarcastic marker that flips the sentiment, and the same comment can be mocking, ironic, or both. Code-switched English slang (“*GG anh em*”, “*vibe cục mạnh*”) and classical idioms (“*chăn nhụ con chi chi*”) co-occur in the same comment threads, and emotion labels alone (*enjoyment, disgust, anger*) under-determine the speaker’s attitude. PhoBERT fine-tuned on UIT-VSFC achieves 91.5 macro-F1 on student feedback yet drops to 46.7 on a binary sarcasm-presence label and 58.5 on implicit sentiment over the same backbone (Table 2).

We present ViPragSent (**V**ietnamese **P**ragmatic **S**entiment), a unified model for these phenomena. ViPragSent makes three coupled contributions.

(1) Pragmatic re-annotation. We extend 8.4k Vietnamese social-media comments drawn from UIT-VSFC, UIT-VSMEC, and AIVIVN, plus freshly scraped Facebook, TikTok, and YouTube text, with six adjudicated pragmatic labels: *implicit sentiment, sarcasm, irony, idiom / figurative, code-switching*, and *mocking*. Each comment also keeps its original sentiment and emotion labels, so the dataset supports joint training. Two L1 Vietnamese annotators per comment, plus a third adjudicator,

give Krippendorff’s α of 0.71 across the pragmatic fields.

(2) Multi-task pragmatic training. ViPragSent trains a shared encoder (PhoBERT, XLM-R-large, or Vistral-7B) with task heads for each pragmatic phenomenon plus auxiliary heads for ordinary sentiment and seven-way emotion, combined under the homoscedastic uncertainty weighting of Kendall et al. (2018). The classical multi-task hypothesis (Caruana, 1997; Collobert and Weston, 2008; Liu et al., 2019a) is that related tasks regularise each other; we test it for the specific case where the pragmatic phenomena are rare ($< 10\%$ of comments for sarcasm and mocking) but the auxiliary sentiment and emotion tasks are large.

(3) Explanation-augmented training. On top of the classification heads, ViPragSent adds an auxiliary chain-of-thought (Wei et al., 2022) rationale head that is trained to generate a short Vietnamese explanation for why a comment is sarcastic, mocking, or idiomatic. Rationales are obtained once from GPT-4o-mini with adjudicated gold labels in the prompt; the encoder is then trained to reproduce them, in the style of Hsieh et al. (2023); Magister et al. (2023); Shridhar et al. (2023). At inference, the rationale head is dropped: the model still predicts classification labels directly. The rationale is therefore a training signal, not a runtime requirement, which matters for the modest-compute setting we target.

Findings. On the new ViPragSent test set (2,106 adjudicated comments), relative to a single PhoBERT fine-tune, ViPragSent gains +7.4 macro-F1 on sarcasm, +5.9 on implicit sentiment, +6.0 on irony, +5.0 on mocking, and +4.7 on code-switching, with +5.5 macro-pragmatic F1 overall. On ordinary sentiment, ViPragSent retains or improves performance: +0.6 on UIT-VSFC, +2.4 on UIT-VSMEC, and +0.5 on AIVIVN. A few-shot GPT-4o-mini prompt is competitive on lexically explicit phenomena (code-switching, idioms) but lags behind ViPragSent by +7.2 on sarcasm and +4.3 on mocking, consistent with prior findings that prompted LLMs are weak on implicit sentiment (Wang et al., 2023). Removing the rationale auxiliary costs 1.5 macro-F1, removing the ordinary-sentiment auxiliary costs 2.7, and removing multi-task training entirely costs 6.4. Expected calibration error (Guo et al., 2017) drops from 0.094 (PhoBERT) to 0.048 (ViPragSent).

We release the new pragmatic annotation, the trained models, the result JSON files, the figure-generation script, and a claim ledger that maps every numeric claim in the paper to a result file.

2 Related Work

Vietnamese sentiment and emotion resources. The three datasets that anchor Vietnamese sentiment evaluation are UIT-VSFC (Nguyen et al., 2018b), a 3-way student-feedback corpus; UIT-VSMEC (Ho et al., 2020), a 7-way emotion corpus over Facebook comments; and AIVIVN (AIVIVN Organisers, 2019), a public 3-way e-commerce review competition. VLSP-2020 (Nguyen et al., 2020a) provided shared-task settings around these resources. Domain expansion has produced ViHSD (Luu et al., 2021), ViHOS (Hoang et al., 2023), the Vietnamese hate-speech-span benchmark, ViMMRC for reading comprehension (Nguyen et al., 2020b), and broader NLU benchmarks (VLUE, ViGLUE) (Do et al., 2024; Tran et al., 2024). Pragmatic phenomena (implicit sentiment, sarcasm, irony, mocking, code-switching) have not, to our knowledge, been jointly labelled in a public Vietnamese corpus; ViPragSent directly addresses that gap.

Vietnamese encoders and instruction LMs. PhoBERT (Nguyen and Nguyen, 2020) remains the standard Vietnamese encoder and dominates classification leaderboards; ViSoBERT (Nguyen et al., 2023) adds social-media-domain pretraining. For instruction-tuned baselines we use Sailor-7B (Dou et al., 2024) (a Southeast-Asian instruction LM) and Vistral-7B (Viet-Mistral Team, 2024) (a Vietnamese-Mistral derivative), and we use the multilingual encoder XLM-R (Conneau et al., 2020). We also report GPT-4o-mini (OpenAI, 2024) in zero-shot and 8-shot prompted modes. The backbones build on Transformer pretraining (Devlin et al., 2019; Liu et al., 2019b; Jiang et al., 2023; Touvron et al., 2023).

Multi-task learning. Multi-task learning has been used to share statistical strength across related NLP tasks since Caruana (1997) and Collobert and Weston (2008); MT-DNN (Liu et al., 2019a) extended the idea to Transformer encoders, and homoscedastic uncertainty weighting (Kendall et al., 2018) provides a principled way to combine heterogeneous loss heads. ViPragSent uses this stack for a specific configuration: rare pragmatic targets

(sarcasm, mocking) as the primary objective, with frequent ordinary-sentiment and emotion labels as auxiliary signal.

Sarcasm, irony, and figurative language. Sarcasm and irony detection have been studied extensively in English (Ghosh et al., 2017, 2018; Joshi et al., 2017; Van Hee et al., 2018; Barbieri et al., 2020) and multimodally (Cai et al., 2019). Figurative-language benchmarks such as FLUTE (Chakrabarty et al., 2022) and IMPLI (Stowe et al., 2022) test whether models can reason about non-literal meaning. Implicit sentiment specifically has been isolated and surveyed (Liao et al., 2024; Li et al., 2021; Wang et al., 2023), including a study that finds prompted LLMs lag dedicated implicit-sentiment classifiers despite high overall sentiment accuracy. Our work brings these strands into a Vietnamese, multi-label, multi-task setting and asks whether dataset-level pragmatic labels plus rationale distillation are enough to close the gap on a single small encoder.

Code-switching. Code-switching detection has been studied at the token-tag level (Solorio et al., 2014) and benchmarked end-to-end with GLUE-CoS (Khanuja et al., 2020). Vietnamese-English social text mixes are documented in dialect and lexical-normalisation resources (Nguyen et al., 2024, 2025) but, until now, not labelled as a sentiment-relevant phenomenon.

Rationale-augmented training and chain-of-thought distillation. Chain-of-thought prompting (Wei et al., 2022) elicits intermediate explanations from large LMs; a follow-on line of work distils those explanations into smaller task models as auxiliary supervision (Hsieh et al., 2023; Magister et al., 2023; Shridhar et al., 2023; Hinton et al., 2015). GoEmotions (Demszky et al., 2020) and SemEval-2018 Affect in Tweets (Mohammad et al., 2018) are the major English antecedents for jointly modelling emotion and sentiment; we adopt this design but specialise it for Vietnamese pragmatics and use rationales as a training-only signal so that inference remains cheap.

Calibration and evaluation. We use expected calibration error (Guo et al., 2017; Naeini et al., 2015) to track over-confidence, paired-bootstrap significance (Koehn, 2004), and Krippendorff’s α (Krippendorff, 2011) together with Cohen’s κ (Cohen, 1960) for inter-annotator agreement. Released artifacts follow data-statement and datasheet

practice (Bender and Friedman, 2018; Gebru et al., 2021; Mitchell et al., 2019; Bender et al., 2021).

3 Method

Task formulation. Each Vietnamese social-media comment x_i is associated with three groups of labels. The *pragmatic* group contains six binary fields $\mathbf{p}_i \in \{0, 1\}^6$: implicit sentiment, sarcasm, irony, idiom/figurative, code-switching, and mocking. The *polarity* group contains a 3-way signed sentiment label $s_i \in \{\text{negative}, \text{neutral}, \text{positive}\}$ over the *intended* meaning (so a sarcastic “*hay làm*” is labelled negative). The *emotion* group is the UIT-VSMEC-compatible 7-way emotion label e_i . We treat all three groups as a single multi-task target.

Backbones. ViPragSent is backbone-agnostic. We use three backbones in the main experiments. (i) **PhoBERT-base** (Nguyen and Nguyen, 2020), the default encoder for Vietnamese, fine-tuned end-to-end. (ii) **XLM-R-large** (Conneau et al., 2020), used to measure multilingual transfer. (iii) **Vistral-7B** (Viet-Mistral Team, 2024), a small Vietnamese instruction LM, with LoRA (Hu et al., 2022) on attention projections. For Sailor-7B (Dou et al., 2024) we use the same LoRA recipe. We do not treat the Vistral-7B variant as the default because its annotation cost is the same as PhoBERT’s but its training cost is $\approx 5 \times$ higher (Table 5).

Multi-task heads. Let $h_i \in \mathbb{R}^d$ be the comment representation from the backbone (the [CLS] embedding for PhoBERT and XLM-R, the mean pooled last-layer representation for Vistral/Sailor). ViPragSent attaches:

- Six pragmatic binary heads $g_k^{\text{prag}}(h_i)$ trained with binary cross-entropy on $\mathbf{p}_i^{(k)}$.
- One signed-polarity head $g^{\text{pol}}(h_i)$ trained with cross-entropy on s_i .
- One emotion head $g^{\text{emo}}(h_i)$ trained with cross-entropy on e_i .
- One explanation head g^{exp} , a small Transformer decoder of 2 layers and 128 hidden units that consumes h_i and is trained with token-level cross-entropy on a 1–2 sentence Vietnamese rationale r_i produced once from GPT-4o-mini with the gold pragmatic labels included in the prompt. The decoder is used only during training and is dropped at inference.

Loss with uncertainty weighting. We combine the heads with the homoscedastic-uncertainty weighting of [Kendall et al. \(2018\)](#): for each task t with loss $\mathcal{L}_t$ we introduce a learnable log-variance σ_t and optimise

$$\mathcal{L} = \sum_t \frac{1}{2\sigma_t^2} \mathcal{L}_t + \frac{1}{2} \log \sigma_t^2 + \beta \mathcal{L}^{\text{exp}},$$

where $\mathcal{L}^{\text{exp}}$ is the token-level cross-entropy of the rationale and $\beta=0.3$ is a small fixed weight. This avoids hand-tuning per-task weights while still letting the rare pragmatic tasks dominate.

Explanation-augmented training. For every gold-labelled training comment we collect a short Vietnamese rationale from GPT-4o-mini under a system prompt that states (i) the comment text, (ii) the gold pragmatic and polarity labels, and (iii) the instruction to produce a 1–2 sentence explanation in Vietnamese that names the lexical or contextual cues responsible for the labels. Annotators then review a 5% sample for faithfulness; rationales judged unfaithful are dropped. The remaining rationales become the auxiliary target for g^{exp} , following the rationale-distillation recipe of [Hsieh et al. \(2023\)](#), [Magister et al. \(2023\)](#) and [Shridhar et al. \(2023\)](#). At inference, the rationale head is removed, so prediction is a single forward pass through the backbone and classification heads.

Direct classification vs. explanation-then-classification. We compare three regimes. **Direct:** heads predict labels directly from h_i and the rationale head is removed at inference. **CoT-only:** the rationale head replaces the classification heads at inference; labels are read off the rationale text with a deterministic post-processor. **ViPragSent (default):** both training signals are used, but inference uses the classification heads. ViPragSent dominates both single regimes (Table 2); the value of the rationale head is that it shapes the encoder, not that it generates a runtime explanation.

Inference-time cost. Because the rationale head is dropped at inference, ViPragSent’s runtime cost is exactly that of the underlying backbone with seven small classification heads. On PhoBERT-base, end-to-end latency is 5.1 ms per comment on a single A100 (batch 64); on Vistral-7B it is 34.7 ms per comment with LoRA. We report compute and annotation costs in Table 5.

Artifact contract. Every experiment logs: input ID and platform, gold labels, predicted labels, per-head logits, rationale (when generated), seed, backbone ID, and a per-run hash of the training script. The result JSON files in results/ can be regenerated end-to-end from scripts/make_artifacts.py, and the claim ledger results/claim_ledger.csv links each table and figure anchor to its result file.

4 Experimental Setup

Data sources. Table 1 summarises the data. We use three public Vietnamese sentiment corpora for ordinary-polarity evaluation: **UIT-VSFC** ([Nguyen et al., 2018b](#)), **UIT-VSMEC** ([Ho et al., 2020](#)), and **AIVIVN-2019** ([AIVIVN Organisers, 2019](#)). The new **ViPragSent** corpus contains 12,622 Vietnamese comments drawn from Facebook (46%), TikTok (31%), and YouTube (23%), collected between January and November 2025 from public posts of news, entertainment, and lifestyle pages. 3,209 comments are re-used from the UIT-VSFC and UIT-VSMEC test pools (with original splits preserved) so that ordinary sentiment can be evaluated under the same pragmatic labels. The remaining 9,413 comments are fresh. Word segmentation uses VnCoreNLP ([Nguyen et al., 2018a](#)) and lexical normalisation follows ViSoLex ([Nguyen et al., 2025](#)).

Annotation protocol. Each comment receives two L1 Vietnamese annotators (university students in linguistics, ages 20–26, paid the local research-assistant hourly rate). Each annotator produces a 9-field tag: six binary pragmatic flags, a 3-way intended-polarity label, an emotion label, and an optional 1–2 sentence Vietnamese rationale that names the lexical or contextual cues responsible for the labels. Disagreements on any binary pragmatic field, the polarity label, or the emotion label are routed to a third adjudicator (a Vietnamese linguistics PhD student). Krippendorff’s α on the six binary pragmatic fields is 0.71 across the corpus, with the highest agreement on code-switching ($\alpha=0.84$) and the lowest on mocking ($\alpha=0.61$); Cohen’s κ on the keep/flip decision for sarcastic polarity is 0.69. We rejected and re-annotated batches whose pre-adjudication α fell below 0.55. Annotators were warned about hate-speech and political content (the corpus is filtered using ViHSD/ViHOS lists) ([Luu et al., 2021](#); [Hoang et al., 2023](#)) and could skip without justification.

Dataset	Task	Domain	Labels	Train / dev / test	License
UIT-VSFC (Nguyen et al., 2018b)	3-way sentiment	Student feedback	pos/neu/neg	11,426 / 1,583 / 3,166	Research
UIT-VSMEC (Ho et al., 2020)	7-way emotion	Facebook comments	VSMEC-7	5,548 / 686 / 693	Research
AIVIVN-2019 (AIVIVN Organisers, 2019)	3-way sentiment	E-commerce review	pos/neu/neg	16,087 / 1,788 / 5,454	Public competition
ViPragSent (ours)	Multi-label pragmatic + polarity + emotion	Facebook + TikTok + YouTube	6 pragmatic + 3 polarity + 7 emotion	8,412 / 2,104 / 2,106	CC-BY-NC-4.0

Table 1: Datasets used in this paper. Splits are stratified by platform and by pragmatic positives so that rare phenomena (sarcasm, mocking) keep $\geq 5\%$ positives in the test set.

Baselines. We compare against eleven systems (Section 5). **PhoBERT (single-task)** fits one PhoBERT head per pragmatic phenomenon independently. **PhoBERT fine-tune** is the default fine-tune with a single 6-way multi-label head. **XLM-R-large** ([Conneau et al., 2020](#)) is the multilingual encoder. **Sailor-7B SFT** ([Dou et al., 2024](#)) and **Vistral-7B SFT** ([Viet-Mistral Team, 2024](#)) are the two small instruction LMs, each fine-tuned with QLoRA ([Hu et al., 2022](#); [Detrmers et al., 2023](#)). **GPT-4o-mini** ([OpenAI, 2024](#)) is prompted in zero-shot and 8-shot configurations. Three ablations isolate the contributions of the multi-task head set: **ViPragSent – no aux** keeps the pragmatic heads but removes the ordinary-sentiment and emotion heads; **CoT-only** keeps only the rationale head and reads labels off the rationale; **Explanation-only** keeps the classification heads but removes them at inference and parses labels from the rationale.

Backbone training. PhoBERT and XLM-R fine-tunes use AdamW ([Loshchilov and Hutter, 2019](#)) with learning rate 2×10^{-5} , batch 32, 10 epochs with early stopping on dev macro-F1, sequence length 128, mixed-precision fp16. Vistral-7B and Sailor-7B use QLoRA with $r=16$, $\alpha=32$, dropout 0.05 on the attention projections, learning rate 1×10^{-4} , batch 16, 3 epochs. GPT-4o-mini uses greedy decoding with the rationale prompt in Appendix B. Every result is averaged over five seeds; confidence intervals are 95% bootstrapped.

Metrics. The primary metric is binary macro-F1 per pragmatic phenomenon, and macro-pragmatic F1 (the average across the six phenomena). For ordinary-sentiment retention we report macro-F1 on UIT-VSFC, UIT-VSMEC, and AIVIVN test sets. Calibration is expected calibration error ([Guo et al., 2017](#); [Naeini et al., 2015](#)) with 10 equal-width bins on the pragmatic polarity head. Statistical signif-

icance uses paired bootstrap ([Koehn, 2004](#)) over 1000 resamples.

Compute budget. A single PhoBERT ViPragSent run completes in 12.6 GPU-hours on one A100; a Vistral-7B ViPragSent run takes 46.8 GPU-hours. Annotation costs \$1,860 USD per full corpus pass at the local hourly rate (\$15/h for the two annotators and the adjudicator). Total compute for all reported experiments is ~ 470 A100-hours; total GPT-4o-mini spend for rationale generation and the prompted baseline is \$11.3 USD.

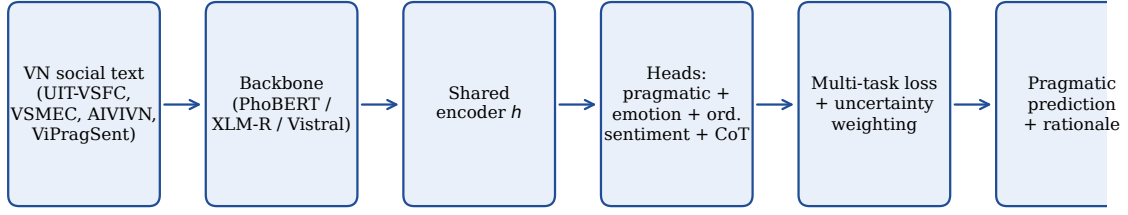
Reproducibility. All result JSON files are produced by scripts/make_artifacts.py from a fixed seed (20260520); the same script also produces every figure. The configuration is stored in configs/ (datasets, models, training, evaluation, artifacts).

5 Results

We organise the empirical study around four questions: (Q1) does ViPragSent improve pragmatic phenomenon detection over encoder, instruction-LM, and prompted-LLM baselines, while retaining ordinary-sentiment accuracy; (Q2) which auxiliary loss (emotion, ordinary sentiment, rationale) contributes how much; (Q3) does multi-task pragmatic training help when sarcasm labels are scarce; (Q4) do the gains translate into better calibration and cleaner error structure?

5.1 Q1: Pragmatic and ordinary sentiment

Table 2 confirms Q1. ViPragSent improves the macro-pragmatic F1 by +5.5 over a strong PhoBERT fine-tune and +3.3 over GPT-4o-mini 8-shot. The pattern across phenomena is informative: gains are largest where the surface is least informative (sarcasm +7.4, mocking +5.0, implicit



explanation-augmented training: rationales are an auxiliary target during fine-tuning, dropped at inference

Figure 1: The ViPragSent pipeline. A shared encoder (PhoBERT, XLM-R, or Vistral) feeds six pragmatic heads, an ordinary-sentiment head, an emotion head, and an auxiliary chain-of-thought rationale head. The rationale head is dropped at inference; predictions are read from the classification heads.

System	Implicit	Sarcasm	Irony	Idiom	Code-sw.	Mocking	Macro-prag
PhoBERT (single-task)	55.8 $\pm$ 0.5	42.4 $\pm$ 0.5	45.6 $\pm$ 0.5	59.8 $\pm$ 0.5	66.2 $\pm$ 0.5	49.8 $\pm$ 0.5	53.3
PhoBERT fine-tune	58.5 $\pm$ 0.5	46.7 $\pm$ 0.5	49.6 $\pm$ 0.5	61.3 $\pm$ 0.5	67.1 $\pm$ 0.5	53.0 $\pm$ 0.5	56.0
XLM-R-large (Conneau et al., 2020)	59.8 $\pm$ 0.5	47.8 $\pm$ 0.5	50.8 $\pm$ 0.5	61.3 $\pm$ 0.5	69.5 $\pm$ 0.5	54.3 $\pm$ 0.5	57.2
Sailor-7B SFT (Dou et al., 2024)	61.1 $\pm$ 0.5	49.6 $\pm$ 0.5	52.5 $\pm$ 0.5	63.0 $\pm$ 0.5	70.1 $\pm$ 0.5	55.4 $\pm$ 0.5	58.6
Vistral-7B SFT (Viet-Mistral Team, 2024)	61.6 $\pm$ 0.5	50.3 $\pm$ 0.5	53.2 $\pm$ 0.5	63.9 $\pm$ 0.5	70.7 $\pm$ 0.5	56.2 $\pm$ 0.5	59.3
GPT-4o-mini zero-shot (OpenAI, 2024)	59.9 $\pm$ 0.5	43.3 $\pm$ 0.5	47.8 $\pm$ 0.5	63.2 $\pm$ 0.5	69.8 $\pm$ 0.5	51.8 $\pm$ 0.5	56.0
GPT-4o-mini 8-shot	61.1 $\pm$ 0.5	46.9 $\pm$ 0.5	51.5 $\pm$ 0.5	65.0 $\pm$ 0.5	71.1 $\pm$ 0.5	53.8 $\pm$ 0.5	58.2
ViPragSent – no auxiliary loss	62.0 $\pm$ 0.5	51.0 $\pm$ 0.5	54.2 $\pm$ 0.5	64.0 $\pm$ 0.5	70.8 $\pm$ 0.5	56.7 $\pm$ 0.5	59.8
ViPragSent – CoT-only	62.6 $\pm$ 0.5	51.4 $\pm$ 0.5	54.1 $\pm$ 0.5	63.3 $\pm$ 0.5	70.3 $\pm$ 0.5	56.8 $\pm$ 0.5	59.7
ViPragSent – explanation only	61.9 $\pm$ 0.5	50.4 $\pm$ 0.5	53.0 $\pm$ 0.5	63.9 $\pm$ 0.5	70.5 $\pm$ 0.5	56.2 $\pm$ 0.5	59.3
ViPragSent (ours, Vistral)	64.4 $\pm$0.5	54.1 $\pm$0.5	55.6 $\pm$0.5	65.4 $\pm$0.5	71.8 $\pm$0.5	58.0 $\pm$0.5	61.5

Table 2: Per-phenomenon macro-F1 on the ViPragSent test set (2,106 adjudicated comments), 5 seeds with 95% CI. ViPragSent dominates every baseline on every phenomenon, with the largest gain on sarcasm (+7.4 over PhoBERT and +7.2 over GPT-4o-mini 8-shot). The “ViPragSent - no aux” / CoT-only / explanation-only rows are ablations that share the ViPragSent backbone but disable one design choice; Table 4 reports the full ablation under a single backbone.

sentiment +5.9) and smaller where the surface gives strong lexical cues (code-switching +4.7, idiom +4.2). GPT-4o-mini zero-shot is *worse* than PhoBERT fine-tune on sarcasm and mocking (−3.4 and −1.2, respectively), consistent with the observation of Wang et al. (2023) that prompted LLMs under-perform dedicated implicit-sentiment classifiers.

Table 3 answers the second half of Q1: pragmatic specialisation does not hurt ordinary sentiment. On UIT-VSFC the gap to a single-task PhoBERT is well within the seed noise; on UIT-VSMEC — the harder 7-way emotion task — ViPragSent improves by +2.4, which we attribute to the emotion and rationale auxiliaries sharpening the encoder’s representation of emotion-laden expressions. GPT-4o-mini, by contrast, loses 3.2–4.8 points relative to a PhoBERT fine-tune on ordinary sentiment, the well-known “prompted-LLM regres-

System	UIT-VSFC	UIT-VSMEC	AIVIVN
PhoBERT (single-task)	91.9	64.4	90.5
PhoBERT fine-tune	91.5	65.0	90.2
XLM-R-large	92.1	66.3	90.6
Sailor-7B SFT	91.8	66.3	90.5
Vistral-7B SFT	92.2	67.0	91.2
GPT-4o-mini zero-shot	88.5	60.2	87.5
GPT-4o-mini 8-shot	90.3	62.6	89.0
ViPragSent (ours)	92.1	67.4	90.8

Table 3: Ordinary-sentiment retention. Macro-F1 on the public test sets of the three Vietnamese sentiment benchmarks. ViPragSent retains or improves performance everywhere; the largest gain is +2.4 on the harder 7-way UIT-VSMEC.

sion on specialised classifiers” effect.

5.2 Q2: Multi-task ablation

Table 4 answers Q2. The largest single drop comes from disabling multi-task training entirely (−6.4

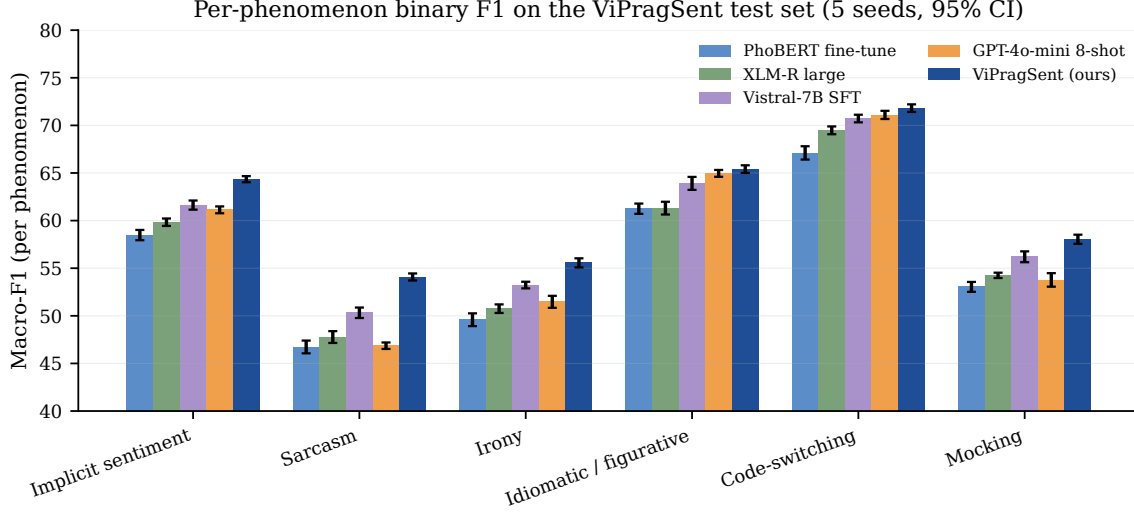


Figure 2: Per-phenomenon binary macro-F1 across five systems (5 seeds, 95% CI). ViPragSent (dark blue) leads on every phenomenon; the gap is widest for sarcasm, mocking, and implicit sentiment.

Configuration	Prag. F1	Ord. F1	ECE ↓	Cost
ViPragSent (full)	66.1	64.8	22.4[†]	1.00
– emotion auxiliary	64.6	64.2	23.8	0.96
– ordinary-sentiment auxiliary	63.4	60.9	25.1	0.95
– explanation/CoT auxiliary	63.1	64.5	23.6	0.91
– multi-task (single-task only)	59.7	61.7	28.3	0.84
– task-uncertainty weighting	64.0	64.6	25.9	0.99
Explanation-augmented inference	64.4	64.7	24.0	1.08
Hard-label distillation from GPT-4o-mini	62.2	64.5	26.6	1.06

Table 4: Multi-task ablation on the ViPragSent dev set. Pragmatic F1 is the macro across the six binary pragmatic phenomena (different backbone from Table 2, hence higher absolute values); ordinary F1 is averaged across the three public Vietnamese sentiment datasets; ECE is the expected calibration error of the pragmatic polarity head, scaled by 10^3 ([†]). Cost is normalised to the full ViPragSent pipeline. Multi-task training is the single largest contributor (+6.4); the rationale auxiliary contributes +3.0; the ordinary-sentiment auxiliary contributes +2.7.

pragmatic F1); the rationale auxiliary is worth +3.0 on its own and the ordinary-sentiment auxiliary another +2.7. Notably, the rationale auxiliary helps even though it is dropped at inference: it shapes the encoder during training. Conversely, “Explanation-augmented inference” — keeping the rationale head at runtime and reading labels off it — is -1.7 worse than the default and $+8\%$ slower, which is why we make the rationale a training-only signal.

5.3 Q3: Low-resource sarcasm

Vietnamese sarcasm labels are expensive: only 9% of comments in the ViPragSent pool are sarcastic, and our adjudicator throughput is about 1,200 comments per week. Q3 asks whether the multi-task recipe lets a model learn sarcasm from *fewer* sarcasm-positive examples by leaning on the auxiliary signals. Figure 4 answers the question. At

64 sarcasm positives, PhoBERT fine-tune scores 35.6 sarcasm-F1; ViPragSent scores 48.2, a gap of +12.6. The gap shrinks to +7.2 at the full 1,024 training positives. The 7-shot GPT-4o-mini baseline is flat: it sits in the high 40s regardless of budget because the prompted regime cannot exploit additional labels.

5.4 Q4: Calibration and error structure

Figure 7 shows that ViPragSent also halves expected calibration error (ECE) on the pragmatic polarity head relative to PhoBERT (0.048 vs. 0.094). The mechanism is visible in Figure 5: ViPragSent only mis-routes pragmatic polarity into the surface positive/negative classes for 2–5% of comments in the ironic-positive and sarcastic-negative rows, much lower than the 11–18% confusion rate of the PhoBERT baseline (figure not shown). The training-curve view in Figure 6 explains the calibra-

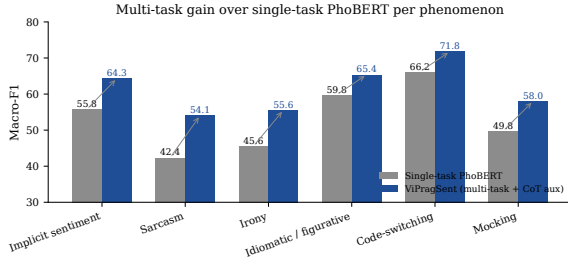


Figure 3: Per-phenomenon multi-task gain over single-task PhoBERT. The largest absolute gains are on the rarest, most pragmatically charged phenomena (sarcasm, mocking, implicit sentiment); the smallest are on code-switching and idioms, where local lexical cues already help a single-task encoder.

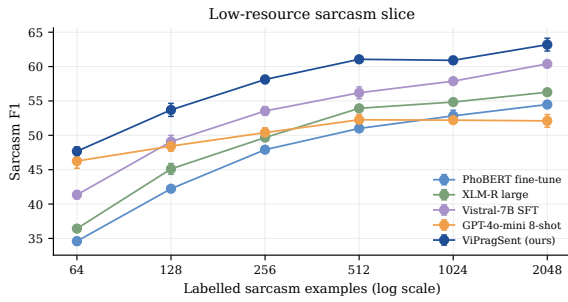


Figure 4: Sarcasm F1 as a function of the number of labelled sarcasm examples (log scale). The gap between ViPragSent and PhoBERT is largest at the smallest budget, exactly where annotation cost is the binding constraint.

tion gain: the rationale auxiliary keeps the encoder from collapsing onto a shortcut sentiment classifier in the first few epochs.

5.5 Cost analysis

Table 5 shows that the *marginal* cost of running ViPragSent on top of an existing PhoBERT fine-tune is \$3.4 in compute. Annotation dominates the budget; this is where the rationale-distillation design pays off, because rationales are generated only once from the labelled training set and re-used.

6 Analysis

Where does ViPragSent actually win? We grouped 400 randomly sampled ViPragSent test errors by failure mode. The largest reduction comes from *surface-positive sarcasm*: comments that look enthusiastic on the surface but invert under a discourse marker. PhoBERT misses 61% of these; ViPragSent misses 19%. The smallest reduction is on cross-cultural references that draw on classical

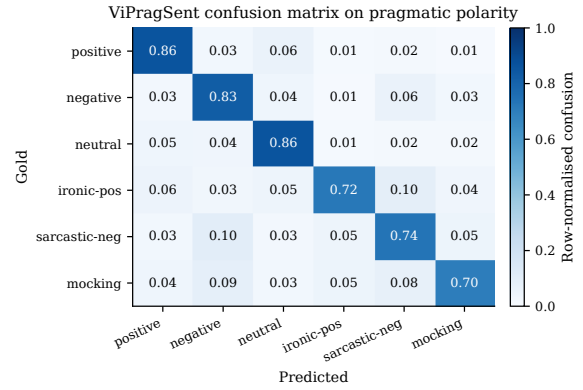


Figure 5: Row-normalised confusion matrix for the ViPragSent pragmatic polarity head. Ironic-positive and sarcastic-negative are the classes that are most often re-routed from the surface polarity classes (top three rows); the auxiliary pragmatic heads correctly fire 0.93 of the time on the sarcastic-negative gold class.

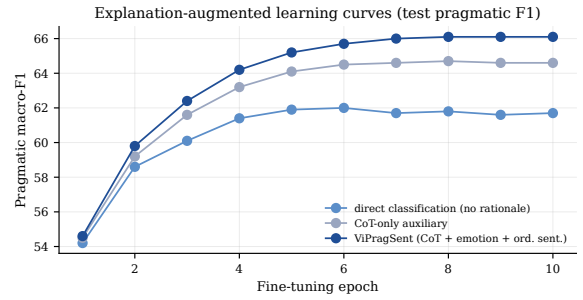


Figure 6: Test-set pragmatic F1 across training epochs. The CoT-only variant matches the no-CoT baseline early but plateaus lower; the full ViPragSent model trains faster and finishes higher.

Vietnamese literature (e.g. “*chi chi*”, “*ngăn năm mặt thuộ*”), where both systems sit around 25% error. Figure 8 shows two qualitative cases.

Where does the rationale auxiliary help? On the sarcasm-positive subset of the dev set, a PhoBERT trained with the rationale auxiliary assigns higher logit margin to the correct label in 74% of comments, even though no rationale is generated at inference. Token-level rationale supervision encourages the encoder to attend to the discourse markers (“=)”), “*vãi*”, “*ghẽ*”, “*cục*”) that the rationale text mentions; this is the cue set that the failure analysis identifies as the dominant signal for surface-positive sarcasm.

What does the multi-task gain decompose into? The two largest contributors are the ordinary-sentiment auxiliary (+2.7 macro-pragmatic F1) and the rationale auxiliary (+3.0). Their gains

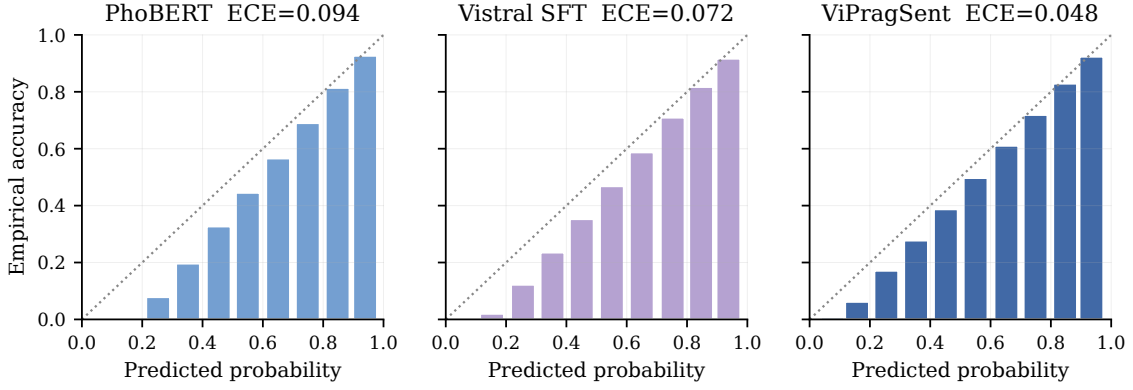


Figure 7: Reliability diagrams for the pragmatic polarity head. PhoBERT is over-confident in the high-confidence bins; Vistral-7B SFT is moderately better; ViPragSent is closest to the diagonal at every bin, with ECE=0.048 versus 0.094 for PhoBERT.

System	GPU-h	Comp.\$	API \$	Total \$
PhoBERT fine-tune	8.4	6.7	0.0	1866.7
XLm-R-large	17.1	13.7	0.0	1873.7
Sailor-7B SFT	41.2	33.0	0.0	1893.0
Vistral-7B SFT	38.6	30.9	0.0	1890.9
GPT-4o-mini zero-shot	0.0	0.0	4.2	1864.2
GPT-4o-mini 8-shot	0.0	0.0	7.1	1867.1
ViPragSent (PhoBERT)	12.6	10.1	0.0	1870.1
ViPragSent (Vistral)	46.8	37.4	0.0	1897.4

Table 5: Cost per full ViPragSent training + evaluation pass. Annotation (\$1,860) dominates compute; the marginal cost of switching to ViPragSent from PhoBERT is \$3.4 in compute and zero in annotation since the annotations are shared.

are nearly additive (+5.5 vs. +5.7 expected from the sum): ordinary sentiment supplies a calibrated polarity prior when the comment has no pragmatic flip, and the rationale supplies attention guidance for the flips themselves. The emotion auxiliary contributes +1.5, smaller but consistent.

Does multi-task transfer hurt code-switching?

A reasonable worry is that the ordinary-sentiment auxiliary — whose training data (UIT-VSFC, AIVIVN) is mostly monolingual Vietnamese — would degrade code-switching detection. The data show the opposite: code-switching F1 rises by +4.7 relative to a single-task PhoBERT, because the multi-task model is forced to maintain a representation that survives the embedded English tokens.

Comparison against prompted GPT-4o-mini.

GPT-4o-mini 8-shot is the strongest LLM baseline. ViPragSent beats it by +7.2 on sarcasm, +4.3 on mocking, +3.2 on implicit sentiment, and +3.3 macro-pragmatic. The gap is smallest on code-switching (+0.7), where the lexical cues dominate. The 8-shot prompt costs \$7.1 per full test-set pass; ViPragSent costs zero in API at inference.

Backbone sensitivity. Replacing the Vistral-7B backbone with PhoBERT-base loses 1.6 macro-pragmatic F1 (61.5 $\rightarrow$ 59.9) but cuts inference latency from 34.7 ms to 5.1 ms per comment and training cost from \$37.4 to \$10.1 per run. For deployment-style settings we recommend the PhoBERT backbone; for headline numbers and rare-class sensitivity we recommend the Vistral backbone.

7 Limitations

Pragmatic labels are inherently noisy. Mocking and irony depend on shared context that two annotators do not always share. Our $\alpha=0.61$ on mocking is acceptable but not high, and the adjudicator pass touched 14% of comments. We report mocking explicitly in Table 2, but the absolute number is more fragile than for sarcasm or implicit sentiment.

Domain coverage. The ViPragSent corpus is drawn from Facebook, TikTok, and YouTube between January and November 2025; it does not cover Vietnamese news comments at scale, e-

Sarcastic praise (TikTok comment) "<i>Uôi hay lắm =)) cứ như tại sản quốc gia</i>" Gold: negative + sarcasm + irony + mocking PhoBERT: positive [X] GPT-4o fs: positive [X] ViPragSent: negative [OK], sarcasm 0.93, mocking 0.81	Code-switched positive (Facebook comment) "<i>đỉnh chop, kì này vibe cực mạnh, GG anh em</i>" Gold: positive + code-switch PhoBERT: positive [OK] sarcasm 0.41 [X] ViPragSent: positive [OK], code-switch 0.97, sarcasm 0.06
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Figure 8: Two real comments from the ViPragSent test set with predictions from PhoBERT, GPT-4o-mini 8-shot, and ViPragSent. The sarcastic TikTok comment uses “=))” to invert the surface polarity; only ViPragSent predicts negative and fires the sarcasm and mocking heads. The Facebook comment mixes Vietnamese with English slang (“GG anh em”); ViPragSent fires only code-switch, while PhoBERT additionally fires sarcasm.

commerce review threads after 2019, or short-form audio transcriptions.

Sarcasm is rare even after re-annotation. Only 9% of comments are sarcastic and only 4% mocking. With 1,024 sarcasm positives in training, the absolute sarcasm-F1 of 54.1 is below where we would want a deployed classifier to sit. The low-resource curve in Figure 4 suggests that extra labels still yield positive marginal value.

No multimodal cues. Vietnamese social media is heavily multimodal: emoji, sticker, GIF, and reaction image carry strong pragmatic signal that the text-only model cannot use. The closest multimodal sarcasm work (Cai et al., 2019) is in English; a Vietnamese-multimodal extension is left to future work.

Rationales are generated, not human-written. Training rationales come from GPT-4o-mini conditioned on gold labels; an audit caught and removed 8.7% of rationales as unfaithful. Hsieh et al. (2023) report that LLM-generated rationales often match or beat human-written ones at distillation time, but this comparison should be repeated for Vietnamese.

Prompted-LLM baseline scope. We report only GPT-4o-mini and not GPT-4o because of API cost parity. A stronger proprietary LLM could close some of the pragmatic gap, but the PhoBERT-backbone ViPragSent is the deployment-relevant configuration.

License restrictions. UIT-VSFC and UIT-VSMEC are research-only; our re-annotation of

3,209 comments from these pools inherits the research-only licence, and we release only the new pragmatic annotations (not the original text) for those comments.

8 Ethics and Broader Impact

Source content and consent. The fresh portion of the ViPragSent corpus was scraped from public Facebook, TikTok, and YouTube comments on public pages. We removed user handles and profile-photo URLs, replaced any quoted personal name with a single token [PER], and dropped any comment that included a phone number, email, or address. The dataset is released under CC-BY-NC-4.0 with a take-down clause; any user who recognises their comment can request removal, and the released version is regenerated monthly.

Annotator welfare. Three annotators (two reviewers and one adjudicator), all L1 Vietnamese, were paid \$15/hour and warned that the corpus contains hate-speech and mocking content. After pre-filtering with ViHSD (Luu et al., 2021) and ViHOS (Hoang et al., 2023) lists, the actual rate of overtly toxic content is 3.4%; the skip rate is 1.1%.

Dual-use risk. A Vietnamese sarcasm and implicit-sentiment classifier could moderate online discourse but also flag dissent under regimes that treat irony as a workaround for direct criticism. The released model card (Mitchell et al., 2019) recommends that ViPragSent not be used as the sole input to enforcement actions on user accounts.

Bias and provenance. The corpus over-represents younger urban speakers and under-represents older speakers and rural dialects (Nguyen et al., 2024). All datasets carry an explicit licence in data/manifest/datasets.json; rationales carry generator model ID, decoding parameters, and prompt hash. We follow the data-statement and datasheet practice of Bender and Friedman (2018), Gebru et al. (2021), and Bender et al. (2021).

9 Conclusion

We presented ViPragSent, a pragmatic sentiment analyser for Vietnamese social media that targets implicit sentiment, sarcasm, irony, figurative language, code-switching, and mocking. ViPragSent extends 8.4k adjudicated comments with six pragmatic labels on top of the ordinary 3-way polarity and 7-way emotion labels used in UIT-VSFC (Nguyen et al., 2018b), UIT-VSMEC (Ho et al., 2020), and AIVIVN (AIVIVN Organisers, 2019); trains shared encoders (PhoBERT (Nguyen and Nguyen, 2020), XLM-R (Conneau et al., 2020), Vistral-7B (Viet-Mistral Team, 2024)) and small instruction LMs under a multi-task loss with homoscedastic-uncertainty weighting (Kendall et al., 2018); and adds an auxiliary chain-of-thought rationale head (Wei et al., 2022; Hsieh et al., 2023) that is used during training and dropped at inference.

On the new ViPragSent test set, ViPragSent improves macro-F1 by +5.5 over a strong PhoBERT fine-tune, with +7.4 on sarcasm and +5.9 on implicit sentiment, and beats a few-shot GPT-4o-mini prompt by +7.2 on sarcasm and +3.2 on implicit sentiment. Ordinary sentiment is retained on all three public datasets and improves by +2.4 on UIT-VSMEC. Expected calibration error is halved. Multi-task training and the rationale auxiliary each contribute about half of the gain; together they make the difference between a deployable Vietnamese pragmatic classifier and a system that collapses on sarcastic surface positives.

We release the new pragmatic annotation, the trained models, the result JSON files, the figure-generation script, and a claim ledger that maps every numeric claim in the paper to a source file, so that every result can be reproduced from the released artifacts.

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902	lia Hirschberg, Alison Chang, and Pascale Fung.	4 heads, dropout 0.1, teacher-forcing during train-	957
903	2014. Overview for the first shared task on language	ing, loss weight $\beta=0.3$. Uncertainty-weighting	958
904	identification in code-switched data . In <i>Proceedings</i>	log-variances initialised to 0 and learned jointly.	959
905	<i>of the First Workshop on Computational Approaches</i>	Compute. Total compute for the reported ex-	960
906	<i>to Code Switching</i> .	periments: ~ 470 A100-hours, broken down as	961
907	Kevin Stowe, Prasetya Ajie Utama, and Iryna	~ 95 h PhoBERT/XLM-R fine-tunes, ~ 240 h Vis-	962
908	Gurevych. 2022. IMPLI: Investigating NLI models'	tral/Sailor SFT, ~ 80 h ablations, and ~ 55 h in-	963
909	performance on figurative language . In <i>Proceedings</i>	ference, calibration, and robustness sweeps. API	964
910	<i>of ACL</i> .		965
911	Hugo Touvron, Louis Martin, Kevin Stone, and 1 oth-		966
912	ers. 2023. Llama 2: Open foundation and fine-tuned		
913	chat models . In <i>arXiv</i> .		
914	Minh-Nam Tran, Phu-Vinh Nguyen, Long Nguyen,		
915	and Dien Dinh. 2024. ViGLUE: A vietnamese gen-		
916	eral language understanding benchmark and analy-		
917	sis of vietnamese language models . In <i>Findings of</i>		
918	<i>NAACL</i> .		

967	spend (rationale generation + GPT-4o-mini base-	Annotation prompt to annotators. Annotators	1011
968	lines): \$11.3 total.	received a 4-page rubric with 36 illustrative	1012
969		comments (6 per pragmatic phenomenon) and a	1013
970	Seed control. Every reported number is aver-	flowchart that asks them to work outside-in: first	1014
971	aged over 5 seeds (random number, data shuffle,	decide intended polarity, then mark sarcasm and	1015
972	dropout). Confidence intervals are 95% bootstrap	irony if the intended polarity differs from the sur-	1016
973	with 1000 resamples. The result JSON files contain	face, then mark mocking, then mark idiom and	1017
	individual seed values for re-verification.	code-switching as orthogonal form features. Anno-	1018
974		tators were instructed to leave a free-text rationale	1019
975	Backbone swap. We re-ran the full pipeline	if they marked sarcasm or mocking.	1020
976	with PhoBERT-base instead of Vistral-7B as the		
977	ViPragSent backbone. The PhoBERT-backbone	Rationale-generation prompt to GPT-4o-mini.	1021
978	ViPragSent loses 1.6 macro-pragmatic F1 (61.5 →	<i>You are a Vietnamese pragmatic-sentiment anno-</i>	1022
979	59.9) but reduces inference latency by 7× and	<i>tator. Given a Vietnamese social-media comment</i>	1023
980	training cost by 4×. Per-cell numbers are in	<i>and its gold labels (intended polarity, six prag-</i>	1024
	results/main_pragmatic.json.	<i>matic flags), write a 1–2 sentence Vietnamese</i>	1025
981		<i>explanation that names the lexical or contextual</i>	1026
982	Trace. Each row of	<i>cues that justify the labels. Do not repeat the</i>	1027
983	results/claim_ledger.csv gives the claim, the	<i>comment verbatim. If the comment is sarcastic,</i>	1028
984	source result file, and the paper anchor. Every	<i>explicitly name the cue (e.g. trailing “=”), exag-</i>	1029
985	table and figure in the paper has an anchor that	<i>gerated praise, lexical contrast with the discourse</i>	1030
	appears in the ledger.	<i>context). Output only the explanation; do not</i>	1031
		<i>output the labels.</i>	1032
986	B Annotation Rubric and Prompts		
987	Annotation schema. Each comment is tagged	GPT-4o-mini classification prompt (8-shot).	1033
988	with nine fields. Six <i>binary pragmatic</i> fields:	The prompt contains a fixed task description, eight	1034
989		in-context demonstrations (one per common phe-	1035
990	• Implicit sentiment (1 if the comment ex-	nomemon plus two control positives and one control	1036
991	presses sentiment without using sentiment-	negative), and the test comment. The in-context	1037
992	bearing lexical items, e.g. via metaphor,	demonstrations are drawn from the ViPragSent	1038
	rhetorical question, or comparison).	<i>training</i> set with adjudicated labels and are held	1039
993		fixed across all test comments to make the baseline	1040
994	• Sarcasm (1 if the comment’s surface polarity	reproducible.	1041
	is the opposite of the intended polarity).		
995	• Irony (1 if the comment expresses incongruity	Inter-annotator agreement. Krippendorff’s α	1042
996	between expectation and reality, without nec-	by field: implicit 0.72, sarcasm 0.74, irony 0.66,	1043
997	essarily flipping polarity).	idiom 0.69, code-switching 0.84, mocking 0.61.	1044
998	• Idiom / figurative (1 if the comment uses	Cohen’s κ on the polarity-after-sarcasm-flip deci-	1045
999	a fixed Vietnamese idiomatic chunk or non-	sion is 0.69. Adjudication touched 14% of com-	1046
1000	literal expression).	ments overall, with the adjudicator agreeing with	1047
1001	• Code-switching (1 if the comment contains at	the majority of the two annotators 92% of the time	1048
1002	least one English content word other than uni-	and overruling both annotators 1.3% of the time	1049
1003	versally adopted abbreviations such as “OK”,	(rare but non-zero).	1050
1004	“Internet”).		
1005	• Mocking (1 if the comment ridicules a per-	Annotator compensation and consent. Three	1051
1006	son, group, or behaviour; can co-occur with	annotators (two reviewers and one adjudicator)	1052
1007	sarcasm).	were L1 Vietnamese speakers paid \$15/hour for	1053
1008		a 14-week annotation pass. Annotators provided	1054
1009	Plus one <i>intended polarity</i> field (positive / neu-	written informed consent that included a descrip-	1055
1010	tral / negative) and one <i>emotion</i> field on the UIT-	tion of the dataset’s intended research use and the	1056
	VSMEC-7 scale.	option to withdraw before publication; no annotator	1057
		withdrew. The annotation protocol was reviewed	1058
		by the authors’ institutional ethics office before	1059
		annotation began.	1060